Lamb, Veal, and Game Products (Food Group 17)

Recently, NDL collaborated with other scientists to obtain analytical data on Australian lamb and veal, as well as New Zealand lamb. NDL also collaborated with Colorado State University on a domestic veal study. Notes on other items in this food group will be included after other studies have been completed.

Australian Lamb and Veal Study

A study was conducted to obtain nutrient values for specific cuts of raw Australian veal and lamb. Australian scientific collaborators sent samples to Texas Tech University, where samples were homogenized, composited, and analyzed for nutrients needed for labeling. The veal cuts analyzed were rib roast, fore shank, and hind shank. Ground lamb sold as 85% lean was also analyzed. Nutrient values were added to SR for each of these cuts.

New Zealand Lamb Study

A study was conducted in collaboration with the New Zealand Meat Industry Association to determine nutrient composition of 25 lamb cuts and offal items (8-10 samples per cut) which are available in retail grocery stores in the US. This study was done along with a study of New Zealand beef cuts, described in the Notes on Beef section of this report. The lamb cuts were boneless chump, hind shank, tunnel boned leg/ chump off/shank off, bone in leg chop/steak, fully frenched rack, partly frenched rack, tenderloin, boneless loin, loin chop, loin saddle, square cut shoulder, boneless rolled netted shoulder, square cut shoulder chops, foreshank, breast, boneless flap, neck chops, ground lamb, liver, kidney, heart, sweetbreads, brains, testes, and swiss cut tongue. These cuts were selected by the members of the New Zealand Meat Industry Association with different items being supplied from different meat plants throughout New Zealand. Retail cuts fabricated from the meat carcasses were prepared for dissection, homogenization and nutrient analysis at Massey University, New Zealand. Weights for component factors such as separable lean, intramuscular and subcutaneous fat (separable fat), bone and connective tissue were determined. Nutrient values were added to SR for each of these cuts in both raw and cooked form (using a braised, fast roasted, fast fried, or boiled method) for “separable lean only” and “separable lean and fat” items.

Retail Veal Study

A study was conducted with Colorado State University (CSU) to obtain nutrient and composition data for 7 representative retail veal cuts. The cuts were: loin chops, loin roast, shoulder blade chops, foreshank (center cut), foreshank (osso buco), cutlets, and ground veal. These retail cuts were obtained from the six major US establishments which conduct their own slaughter of special fed (non-bob veal) US calves. The locations were Greeley CO, Collingswood NJ, Detroit MI, Harleysville PA, Franklin WI, and Vineland NJ. Raw and cooked samples (n=6 per cut) were dissected using

standard protocols and then homogenized, composited, and analyzed at CSU for proximates, fatty acids, cholesterol, and ICP minerals. The B-vitamins, vitamins D3 and 25(OH)D, and selenium were analyzed at a validated commercial laboratory. Choline and vitamin E were analyzed at specialized laboratories. Weights of component factors for each cut, such as separable lean, separable fat, and bone and connective tissue, were determined. Nutrient values were added to SR for these cuts in raw and cooked form (grilled, braised or pan-fried) for “separable lean only” and “separable lean and fat”.

Legumes and Legume Products (Food Group 16)

The legumes included in this food group are restricted to the mature, dry seeds of the family Fabaceae or Leguminosae, as well as products made from them. The immature seeds or pods and other parts of the plant, such as leaves, tubers, and sprouted seeds are included in Food Group 11, Vegetables and Vegetable Products.

The sources of nutrient data include the scientific literature, analytical studies conducted by NDL and other government agencies, and the food industry. Since the inception of NFNAP (p. 52) in 1997, a few legumes and legume products (common beans, baked beans, refried beans, chickpeas, peanut butter, and tofu,) have been sampled and analyzed. Data for commonly consumed raw and cooked legumes were obtained by a comprehensive study conducted in the 1980's at the University of Idaho. Data on other raw and cooked legumes, for the most part, have been obtained from the scientific literature. In most cases, data for other processed legumes were supplied by the food industry or obtained from the scientific literature.

Data are presented for raw, cooked, and canned legumes. Legume products, such as peanut butter, soy milk, soy flour, isolates and concentrates, tofu, tempeh, and natto, are also included in this food group. If appropriate, data are presented for both the unprepared and the prepared forms of the food.

Nutrient data for different forms of the same legumes and the legume products were not necessarily derived from the same sample. That is, a single sample of kidney beans was not analyzed in all forms given in the database: raw, cooked, and canned. The data were obtained from many sources and most likely represent different crop years, growing areas, cultivars, processing techniques, lengths and conditions of storage, and possibly different methods of analysis. Therefore, in a comparison of different forms of a legume, nutritional differences should not be ascribed solely to the effect of processing or preparation methods.

Raw legumes.--Although nutrient data were available for several cultivars of some legumes, the data base for any one cultivar was too small to justify giving separate entries by cultivar. Mature legumes are not eaten raw because toxic factors, such as hemaglutinins and trypsin inhibitors (Akroyd and Doughty, 1982), are present. Raw, mature legumes are also unpalatable and hard to digest. The usual processing or preparation methods, such as cooking and canning, remove or inactivate these toxic factors. Cooking or other processing of legumes also makes them more palatable by reducing the beany flavor that some find objectionable.

Prepared legumes.--Nutrient data for cooked legumes were sometimes unavailable or incomplete. In these cases, nutrient values for the cooked form were calculated from data for the raw form of the same legume. For example, nutrient data for boiled peanuts were calculated from data for raw peanuts.

Appropriate true nutrient retention factors were used to calculate the nutrient content of the cooked foods after adjusting for moisture changes. These are presented in the USDA Table of Nutrient Retention Factors, Release 6 (2007). The percentage yields for cooked legumes prepared from raw legumes are given in Table 22. The increased moisture content for cooked legumes leads to the average yield factor of about 2.5.

Table 22. Yields Factors for Selected Cooked Legumes

Legume	Yield Factor Ratio of Weights[1]
Beans:	
Black	2.3
Cranberry	2.3
Great northern	2.5
Navy	2.3
Pink	2.3
Pinto	2.4
Red kidney	2.4
Small red	2.3
Small white	2.3
Broadbeans	2.8
Chickpeas	2.1
Cowpeas	2.6
Lima beans:	
Baby	2.4
Large	2.6
Lentils	2.7
Mung beans	3.2
Peas, split	2.5
Pigeon peas	2.6

[1] Ratio of Weights = (Weight of legume cooked / Weight of legume, uncooked)

Sodium values for cooked legumes are relatively low because no salt was added. Since the sodium content of tap water varies according to location (0-39 mg/100g [NDB No. 14429]), the sodium value of cooked legumes may be underestimated if the water supply naturally contains significant amounts of sodium. Sodium content of cooked legumes will depend on the amount of salt used in cooking and can be as high as that of canned legumes. Sodium values for cooked legumes with added salt were calculated by adding the sodium content of approximately 1/8 of a teaspoon of salt per 100 grams of legumes or 236 mg of sodium to the sodium naturally occurring in the cooked vegetable with no salt added. Tap water also contains other dissolved minerals. The content varies considerably depending on the source and treatment of the local water supply.

Retention values are generally based on cooking methods that minimize the loss of nutrients, particularly the water-soluble vitamins, primarily due to leaching into the cooking water. Nutrient values of cooked legumes obtained by these procedures tend to be higher than those values for the same legumes cooked by less-than-optimum methods. Some conditions that affect the retention of nutrients in legumes include cooking method, size and shape of the legumes or amount of surface area, maturity, condition of the legume, proportion of broken seeds, amount of cooling water and cooking time.

Nutrient values for multi-ingredient mixtures such as home-prepared Boston baked beans were calculated from recipes developed by the US Department of Agriculture. Values for each nutrient provided by the ingredients used in the recipe were totaled. Nutrient values were adjusted by using appropriate nutrient retention values to account for any changes caused by evaporation or hydration and heat destruction due to cooking procedures.

Data for canned legumes were often developed for the purpose of nutritional labeling; therefore data are presented for the total can contents. During the canning process, and sometimes during cooking, the cotyledons of legumes rupture, releasing starch into the brine; therefore, draining of the liquid medium is difficult. However, cooks do generally drain these products. Some packers may add sugar to certain canned legumes and this may also affect the nutrient content.

Nomenclature.--To aid in identifying individual legumes listed in the tables, the scientific name of the legume is included in the food description file, usually on the raw form of the legume. The USDA Germplasm Resources Information Network (GRIN), (USDA, 2011) was used as the basic reference for the scientific and preferred common names.

Identifying legumes by their common names, however, is often confusing because these names are not always applied to the same food in different geographical locations. Some names of legumes in common use or unique to one region of the country have been included in the common name filed of the food description file. Additional descriptions of legumes and legume products are provided in the following paragraphs.

In other countries and among immigrants to the United States, various terms are often substituted for “legume.” The term “pulse” is sometimes used for a legume having a low fat content. Pulses include common beans, broadbeans, peas, and lentils. Soybeans and peanuts are sometimes referred to as leguminous oilseeds (Akroyd and Doughty, 1982). Data on oils derived from these sources, are found in Food Group 4, Fats and Oils.

“Dhal” or “dal” are legumes that have been dehulled and split. This practice is common in India to shorten the cooking time. “Gram” is a term sometimes used in India for the whole seed of any legume (Yamaguchi, 1983) and sometimes used as another name

for the chickpea (Duke, 1981). There are a number of other "grams." For example, red gram refers to pigeon peas, green gram to mung beans, and Bengal gram to chickpeas.

Adzuki beans (*Vigna angularis*) are grown primarily in East Asia, but have been introduced into the southern United States and Hawaii. These beans are eaten either boiled or fried. Adzuki beans ground into a fine paste are used in some confections, such as yokan. Canned adzuki beans, sweetened with sugar, are commonly sold in Hawaii (Duke, 1981).

Common beans (*Phaseolus vulgaris*) are native to the tropical areas of Central and South America. They include black, black turtle soup, cranberry, French, great northern, kidney, navy, pink, pinto, and white beans, which are widely grown in many areas around the world. White beans are commonly used to prepare many types of baked beans. Pinto beans are used in many Mexican recipes (Akroyd and Doughty, 1982, Duke, 1981). Data were combined for several types of kidney beans to produce an overall figure for kidney beans.

The broadbean (*Vicia faba*) has been cultivated in the Mediterranean region and the Near East since ancient times. It was the only bean known to Europeans until the common bean (*P. vulgaris*) was introduced from the New World. There are two major subspecies of broadbeans. Var. major, which has large flat seeds, is usually consumed by humans. Var. equina, which has small, globular seeds and commonly called field bean or horse bean, is used for feeding livestock. Broadbeans are used in many dishes in the Mediterranean region, such as falafel (Akroyd and Doughty, 1982).

Carob flour or powder (*Ceratonia siliqua*), also called St. John's bread, may be used in some foods as a chocolate replacement. The seeds are ground to extract a gum, known as locust bean gum, which is used in many food and industrial products. The pods are ground to produce carob flour. Vegetable oil or other fats are frequently added to the raw carob flour to make confectionary coatings or candy bars. The carob tree is native to the eastern Mediterranean region and has been introduced to California and other areas (Akroyd and Doughty, 1982, Duke, 1981).

Chickpeas (*Cicer arietinum*) or garbanzo beans are one of the most commonly consumed legumes in India and in the Middle East (Akroyd and Doughty, 1982). In the United States, canned or cooked chickpeas are a common item at many salad bars. In the Middle East chickpeas are used in many dishes such as hummus and falafel. These items have become more popular in the United States.

Cowpeas (*Vigna unguiculata*), or black-eyed peas, are cultivated in the southern United States and in many tropical areas. There are three major subspecies: *Vigna unguiculata unguiculata* is the common cowpea or black-eyed pea; *Vigna unguiculata cylindrica*, or catjang, is used whole or split but is more frequently used as forage; *Vigna unguiculata sesquipedalis*, or yardlong bean, has pods that may grow to 36 inches in length and is commonly used in Asian cooking as a vegetable. The mature seeds are also used (Akroyd and Doughty, 1982).

Hyacinth beans (*Dolichos purpurens*), also known as lablab, are native to Asia and have been cultivated in India for centuries. The mature seeds are eaten as a dahl.

Lentils (*Lens culinaris*) originated in the Mediterranean area. The seeds are usually boiled and served in soups and stews (Akroyd and Doughty, 1982).

Lima beans (*Phaseolus lunatus*) originated in tropical regions of the Americas and are now grown in tropical and subtropical areas around the world. Baby lima beans grow in most areas of the United States (Duke, 1981). There are two major subgroupings of lima beans: the small or baby type and the large lima beans.

Lupins (*Lupinus* spp.) are found in the Americas and in the Mediterranean region. There are four major species. White or Egyptian lupins (*Lupinus albus*) are common in the Mediterranean region and were cultivated by the Romans. Seeds are treated by soaking, then boiling, and sometimes additional soaking. Sweet strains with less alkaloids have been developed.

Blue lupin (*Lupinus angustifolius*) originated in northern Europe and is grown primarily for animal feed. Yellow lupin (*Lupinus luteus*) is native to southern Europe and the Mediterranean. Low-alkaloid varieties have been developed. Tarwi or pearl lupin (*Lupinus mutabilis*) has been grown in South America for centuries. Special preparation methods are required to remove the alkaloids. Low-alkaloid types are being developed (Akroyd and Doughty, 1982, Duke, 1981). Because of limited data, nutrient values for all four subspecies have been combined in the tables.

Mothbeans (*Vigna aconitifolia*) are native to India and are eaten whole or as a dhal. The seeds are also used as a source of flour (Duke, 1981).

Mung beans (*Vigna radiata*), also called green gram in India, are native to tropical areas of Asia and are widely grown there. Recently, mung beans have been introduced to the United States. In China and the United States mung beans are commonly grown for sprouting and are consumed as a vegetable. The mature seeds can be boiled and eaten. They can also be ground into a flour for use in bakery products and fried snack foods (Akroyd and Doughty, 1982). Mung beans are also made into a noodle-like product called long rice. A similar product made from mung bean flour is cellophane noodles.

Mungo beans (*Vigna mungo*), sometimes called black gram, originated in India and are also grown in the West Indies (Vaughan and Geisler, 1997). Mungo beans are eaten either whole or as a dhal. They also can be boiled or roasted and ground into flour for use in cakes and breads.

Peas (*Pisum sativum*), or field peas, originated in southwest Asia and are now grown in temperate areas around the world (Akroyd and Doughty, 1982). They were once named as different species-—garden peas and field peas—but are now classified

together. Field peas are hardier, have smaller seeds, and are usually grown for the mature seeds.

Peanuts (*Arachis hypogaea*), native to Latin America, are now grown in tropic, subtropic and warm-temperate areas of the world. In the United States over 60 percent of the peanuts are processed into peanut butter, about 20 percent are roasted, and about 19 percent are used in confections (USDA-ERS, 2014). In other countries, peanuts are produced primarily for their oil and the remaining peanut cake is used for livestock feed. The three main types of peanuts are Virginia, Spanish, and Valencia. Virginia peanuts have large seeds and usually contain two seeds per pod. Spanish peanuts have small seeds and their pods also contain two seeds. Valencia peanuts also have small seeds and the pods contain two to five seeds (Brouk, 1975). Nutrient data on the different cultivars of peanuts were combined to generate overall values for peanuts: Virginia (including runner), 90.6 percent; Spanish type, 8.5 percent; and Valencia type, 0.9 percent. By federal regulation (21 CFR 164.150), peanut butter must contain at least 90 percent peanuts, and not more than 10 percent seasonings (including sugar, salt, and oil), and stabilizing ingredients.

Pigeon peas (*Cajanus cajan*), or red gram, were probably native to Africa, spreading in prehistoric times to Asia (Vaughan and Geisler, 1997). In India, pigeon peas are usually consumed as a dahl.

Soybeans (*Glycine max*) are among the most important sources of protein and oil known to man. Indigenous to eastern Asia, where they have been used in myriad ways for centuries, soybeans are now cultivated in eastern and southeastern Asia as well as in the Americas, predominately in the United States and Brazil (Akroyd and Doughty, 1982, Duke, 1981, Vaughan and Geisler, 1997). A number of fermented soybean products are known in east and southeastern Asian countries, and in recent years these have attracted a following in the United States.

Shoyu (Japanese soy sauce) is made from equal parts soybeans and cracked, roasted wheat, plus salt and water. The mixture is inoculated with *Aspergillus soyae* mold and fermented from 6 months to as long as 5 years. Tamari is a different product made with little or no wheat. In the United States a non-fermented, synthetic product known as soy sauce is prepared from hydrolyzed soy protein, caramel coloring, corn syrup, salt, and water (Shurtleff and Aoyagi, 1979a) rather than using the traditional fermentation method. Soy sauces are commonly used as condiments in east and southeastern Asian style cooking. It is also used in western cuisines and prepared foods.

Miso, or soy paste, is made from soybeans, a grain (either rice or barley), salt, and water. The mold *Aspergillus oryzae* is introduced for fermentation (Shurtleff and Aoyagi, 1976). Many different types of miso are marketed and rarely are identified in the scientific literature. Data for a number of market samples have been combined in the database, which explains the somewhat large variation in some nutrient values.

Natto is made from whole, cooked soybeans, which are inoculated with the bacterium *Bacillus subtiis*. Natto is often served over rice or noodles as a main dish or used in soups and salads (Shurtleff and Aoyagi, 1979a).

Tempeh, from Indonesia, is made from cooked soybeans bound together with the mycellia of the mold *Rhizopus mycellius*. The product is made into cakes or patties and often sliced and fried (Shurtleff and Aoyagi, 1979b).

Tofu, another soy product, is prepared by precipitating the protein of soy milk with any of several coagulants. Tofu is prepared by soaking the whole soybeans overnight and later grinding them with water before draining. The resulting soy milk is pressed from the cooked, ground soybeans, leaving a white or yellowish pulp consisting of the insoluble parts of the soybean. This pulp is known as okara, and can be used in many recipes (Shurtleff and Aoyagi, 1979a). Any of several coagulants is then added to precipitate the protein and form the curds. Nigari, the traditional coagulant used in Japan, contains primarily magnesium chloride. Calcium chloride, calcium sulfate, seawater, lemon juice and vinegar can also be used. As expected, the composition of the coagulant affects the calcium and magnesium content of the finished product. Excess liquid is pressed from the curd, which in turn affects the firmness of the tofu (Shurtleff and Aoyagi, 1979a). Tofu is available in a number of "firmness" types, i.e. soft, silken, medium, firm, extra firm, and others reflecting the amount of water pressed from the curd. The amount of water pressed from the tofu, reflected in the firmness term used, will also have concomitant effect on the nutrient content—firm tofu with less water will have a higher nutrient content than soft types with more water. However, there is no standard for these terms, and one company's tofu using a particular term, may be more similar to another company's product using a different term.

Soymilk is a beverage produced commercially from soybeans. In the United States, soy milk may be used by individuals who choose not to consume animal products, are allergic to cow's milk, or who are lactose intolerant. Some infant formulas are based on soy milk. Soy milk can be processed in many of the same ways as cow's milk and can be substituted for it in many recipes.

In the United States and in other countries, soybeans are utilized as a source of oil, and the resulting defatted meal was formerly used for animal feed. In recent years, however, the defatted soy meal has been used in the preparation of many soy-based products. Soy flour and soy grits can also be prepared from the defatted meal. Soy flour is used in many foods as is or may be extruded into various soy-based products. Soy protein concentrates are processed to remove most of the non-protein compounds, primarily soluble sugars, from the defatted soy flour by wet extraction. Concentrates are often extruded in the preparation of many products.

Soy protein isolates have had nearly all the non-protein constituents removed. The soy extract can be either spray-dried or extruded into an acid medium to form fibers resembling meat, which are marketed as textured vegetable protein. Soy protein

isolates are also used as an ingredient in a number of food products, both as a protein extender and for their functional properties.

Modern processing methods have been used experimentally on a number of other legumes, but none have reached large—scale commercial production or have gained the commercial acceptance of soy products.

Winged beans (*Psophocarpus tetragonolobus*) are native to southeast Asia and have been introduced to tropical areas of the United States, such as Hawaii, Puerto Rico, and southern Florida. The pods, leaves, stems, and tubers of this plant are all edible and are included in Food Group 11, Vegetable and Vegetable Products. Only data on the mature seeds, however, are reported here. The mature seeds can be steamed, boiled, roasted, fermented, or processed into milk or into products such as tofu or tempeh (BOSTID, 1981).

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